SkillNavora · PM Master Handbook

MODULE 7 · DETAILED CHAPTER

Product Analytics and Metrics

How product managers define success, track it every day, and turn numbers into decisions.

Based on the Week 6 Analytics and Metrics class (Product Management Launchpad). Written by Varun Kumar.

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PART 1 · HOW PRODUCT MANAGERS DEFINE SUCCESS

1. What a metric is

A metric is a quantifiable measure used to assess success, or failure, against a goal. The metrics that give the most insight are called KPIs (Key Performance Indicators).

Anatomy of a metric

Part	What it means	Example
Definition	What the metric tries to measure	Count of users, depth of engagement, degree of retention
Value	The numerical value	12,400 weekly active users
Time grain	The period the measurement covers	Monthly vs weekly active users; week-4 vs week-on-week retention
Cohort	The slice of users being measured	New users vs existing users

A metric without a time grain and a cohort is ambiguous. "Retention is 30%" means little until you know which users, measured when.

QUOTE TO REMEMBER

"You can't manage what you don't measure." (Peter F. Drucker)

2. Why metrics matter

Metrics turn product work into clear decisions. They do four jobs in a loop:

Track See what is happening.

Diagnose Understand why it happened.

Decide Choose what to improve.

Align Show progress to the team.

Without metrics, opinions dominate. With metrics, patterns become visible. The good PM habit is simple: **measure before you change**.

3. From user value to business value

Good product metrics connect what users are trying to achieve with what the business is trying to grow. Think of it as a bridge with three steps:

1. **User job:** the outcome the user wants (for example, "get set up and go live faster").
2. **User value metric:** a measure that the user job is working (for example, time to first completed workflow).
3. **Business value metric:** a measure of business impact (for example, activation rate, expansion potential, lower churn risk, revenue).

Example: a B2B onboarding platform. Faster setup leads to a shorter time-to-value, which leads to renewal confidence.

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4. The mental model for product metrics

The goal of every product is to **solve customer problems** and **generate value for the business**. Metrics must cover both sides.

CUSTOMER SIDE

1. **Identify all users and stakeholders.** One product can have more than one type of user.
2. **Identify their Jobs To Be Done (JTBD).** The higher-order problem the user wants solved.
3. **Define a success metric** that captures completion of the JTBD. Often one or more metrics capture true success for the user.

BUSINESS SIDE

1. **Identify the business goal.** For-profit companies ultimately aim to generate shareholder value. Vision and mission statements also make excellent goal statements.
2. **Define a success metric** that shows the product supports the business model.
3. **Add guardrail (counter) metrics** to guard against side effects.

Worked example: a goal that is hard to measure

The class example starts with "to lead a successful life". The metrics are a set, not one number:

Area	Metric
Financial health	Net worth
Physical health	Set of biomarkers (for example HbA1c, lipid panel)
Mental health	Cortisol level
Social health	Number of meaningful relationships

Two lessons follow. Defining success is hard when the goal is ambiguous. And you usually need many metrics to capture success.

5. Exercise: Instagram Reels

Work through the three stakeholders of a short-video feature:

Stakeholder	JTBD	Success metric (JTBD completion)
Content creators and influencers	Build a fan following or a captive audience	Total user reach, total followers, total creates
Content consumers	Find inspiring, relatable or funny content	Total watch time, total likes, comments and shares
Advertisers and brands	Influence targeted users to buy	Total ad impressions, clicks, purchases

The business-goal metrics for the same product: **revenue**, **number of new users added**, and a **guardrail** such as a drop in overall use of the app.

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6. The ladder of metrics

Metrics come in categories. Knowing the category tells you what you can control, when you will see the signal, and how it connects to the result.

North Star Metric (NSM)

The single metric, sometimes called the One Metric That Matters (OMTM), that signifies that the product is delivering value. Examples from the class:

- **Swiggy:** number of successful orders delivered.

- **Airbnb:** number of bookings (and successful stays).
- **Spotify:** total listen time, supported by a "magic moment" (users who follow 10 artists in the first week).

Build the ladder under the North Star

The Airbnb example in the class builds a ladder under its North Star:

Rung	Type	Airbnb example
North Star	The one action that creates value	Number of bookings (and successful stays)
Input	What we control	Number of high-quality listings added
Output	What we get	Number of bookings made
Leading	Early signal, predicts the future	Number of views on listings
Lagging	Confirms the past	Revenue, e.g. commission per booking

Spotify in practice

Leading metrics (# of app installs, monthly active users, # of playlist creations, efficacy of the recommendation system) feed the North Star of total listen time. The magic moment (following 10 artists in the first week) sits upstream of listening. Lagging metrics (subscription revenue, ad revenue, other revenue) confirm the outcome.

THE RULE OF THUMB

If inputs improve, outputs usually follow. Leading metrics tell you where you are going. Lagging metrics tell you where you reached.

Input and output, in one line each

- **Input metrics** measure the controllables you can directly influence, such as the number of user invites sent.
- **Output metrics** measure the outcomes that inputs cause, such as total sign-ups from user invitations.

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7. Leading vs lagging, input vs output: the 2×2

Put the two axes together. Controllability (can the team directly move it?) and temporality (does it predict the future or report the past?) give four quadrants.

LEADING INPUTS (HIGH CONTROL, PREDICTIVE)

The daily levers product and growth teams control. Slack: number of team-member invites sent during onboarding. SaaS CRM: active CRM integrations connected on Day 0. E-commerce: autocomplete latency and zero-result query rate.

LEADING OUTPUTS (LOW CONTROL, PREDICTIVE)

Early customer behaviour that signals long-term value. Slack: a team reaching 2,000 sent messages (the magic moment). Spotify: user saving 10+ songs and following 3 artists in week 1. Duolingo: day-7 streak completion rate.

LAGGING INPUTS (HIGH CONTROL, RETROSPECTIVE)

Actions the organisation completed in the past. Engineering: story points shipped, PRs merged, sprint velocity. Growth: total paid marketing budget deployed. Content: help-centre articles

LAGGING OUTPUTS (LOW CONTROL, RETROSPECTIVE)

The business and financial outcomes. Annual recurring revenue (ARR), net dollar retention (NDR), customer lifetime value (LTV), LTV:CAC

published this quarter. These measure effort, not customer value.

ratio, quarterly EBITDA. These are the scoreboard. You cannot operate them directly.

8. Build a metric system, not one metric

A single metric rarely tells the full story. A complete system has five layers. Example: a B2B collaboration product.

Layer	Question it answers	Example
North Star	What long-term value are we creating?	Weekly active teams
Input metrics	What behaviours drive the result?	Invites sent, workflows created
Output metrics	What moved after launch?	Activation rate, workflow completion
Guardrails	What should not get worse?	Error rate, latency
Diagnostic metrics	Where exactly is the problem?	Drop-off after the invite step

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9. B2B vs B2C metrics

A product can look similar in both models, but the metric logic is often different.

	B2C (consumer)	B2B (business)
Unit of behaviour	User-level behaviour	Account or team-level behaviour
Decision	Shorter decision cycle	Multi-stakeholder adoption, longer sales and renewal cycle
Focus	Conversion and retention	Activation by account, seat adoption, churn risk, expansion revenue
Example metrics	DAU, checkout conversion, repeat rate	Account activation rate, seats adopted, renewal health

10. B2B health dashboard and funnel analysis

A B2B PM needs a dashboard that shows **value**, **risk** and **growth** at the account level:

Lens	Question	Metrics on the dashboard
Value	Are accounts getting live and using the product?	Activation rate, weekly active accounts, adoption breadth, time-to-value
Risk	Which accounts are slipping, blocked or disengaging?	Renewal health, health score distribution, top at-risk accounts
Growth	Which accounts are ready to expand?	Expansion potential, expansion revenue

Funnel for B2B activation

A funnel shows where accounts drop before they become active. The class example:

Step	Share of accounts	What the drop usually means
1. Account created	100%	Starting point
2. Workspace connected	74%	Setup friction
3. First teammate invited	48%	Collaboration value unclear
4. First workflow completed	31%	Product complexity too high
5. Weekly active team	22%	Habit not formed yet

Funnel analysis answers three questions: where users drop, which step blocks value, and what to test next. The biggest leak is usually the best experiment to run.

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11. Worked activity: metrics for Google Maps

The class asked students to do three things: identify the users and their JTBD, define a metric for JTBD completion and a metric for business support, then build a metric ladder with leading, lagging, input and output relationships.

	Drivers and commuters	Local businesses	Google platform
Core JTBD	Find the fastest, safest, congestion-free route with a reliable arrival ETA	Turn nearby foot traffic into verified paying customers	Build the world's most trusted spatial map and monetise high-intent commercial search sustainably
North Star	% of trips taken on optimal routes (time, road condition, traffic, tolls)	Place-sheet actions: calls, direction clicks, table reservations, menu views	Net high-intent monetisation (local search ads plus Maps platform API revenue)
Leading	# of active navigation sessions started	# of claimed business profiles, updated menus	Sponsored pin impressions, API query volume
Lagging	Weekly active drivers, 90-day app retention	Merchant ad spend, quarterly merchant retention	Consolidated ads and enterprise recurring revenue
Input → output	Reroute latency (under 2 s), hazard reporting speed → % completed optimal trips, trip NPS	Profile completeness, review response rate → place-sheet views, in-store visits	Sponsored pin context relevance → ad click-through rate, enterprise renewals
Guardrails	Trip abandonment rate (under 2%), ETA error (under 3 min), distracted-driving alerts	Fake or duplicate listing rate (under 0.1%), spam review volume, dispute resolution time	Map clutter index, navigation task completion drops, battery drain, privacy opt-outs

WHY THIS MATTERS

Each stakeholder gets its own JTBD and metric, but all of them connect to one business metric. The guardrails are what stop one side from winning at the other's expense. For example, pushing more sponsored pins could hurt the driver's trip NPS.

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12. Goodhart's Law and metric gaming

THE PRINCIPLE

"When a measure becomes a target, it ceases to be a good measure." (Charles Goodhart, economist. Marilyn Strathern's later formulation is the version most often quoted.)

How metrics get gamed

- **Proxy displacement.** Teams confuse the proxy (clicks, raw DAU) with the real value (completing the core workflow).

- **Path of least resistance.** Under sprint or OKR pressure, teams optimise the easiest way to move the number, not the hardest way to create value.
- **Compounding side-effects.** Unmonitored dimensions degrade silently until trust, retention or unit economics collapse.

Real PM traps from the class

Trap	What happens
Notification trap	Sending five promotional push alerts boosts DAU by 8% in the short term. App uninstalls and muted alerts then rise sharply.
Signup friction trap	Pre-checking a newsletter box lifts registration conversion. It produces low-intent churn and spam complaints.
Feature tour trap	Forcing five unskippable walkthroughs claims 100% feature adoption. Users click "Next" without learning anything.

The senior PM defence

- **Paired counter-metrics.** Never judge an output metric alone. Pair volume with quality, retention and friction guardrails.
- **Holdout groups.** Keep a permanent holdout (for example 5% of users) to prove true incrementality against what would have happened anyway.
- **Qualitative triangulation.** Pair the dashboard with weekly customer interviews and support tickets to catch metric distortion early.

13. Counter-metrics and guardrails

Every primary metric should be balanced by a guardrail that protects the user or the platform. The class architecture:

Focus area	Primary metric (what we push)	Gaming risk (what can break)	Counter guardrail (what we defend)
User acquisition	New sign-ups and app installs; blended and paid CAC	Low-intent click-farm traffic, bot sign-ups, deceptive ad conversions	Day-30 retention at or above 25%; chargeback and fraud rate under 0.1%
Engagement and growth	DAU; push alert open rate and CTR	Over-notifying users to create fake DAU spikes; annoyance, muting, churn	App uninstalls per 1K sessions under 1.5%; notification mute or opt-out rate
Monetisation and pricing	ARPU; paywall conversion and upsell rate	Dark patterns at checkout, hidden auto-renewals, aggressive paywalls	Support-ticket ratio under 2%; 30-day refund and dispute volume
Platform reliability	Feature release velocity; experiment concurrency	Shipping unvalidated code, bloated dependencies, latency	P99 end-to-end latency under 400 ms; crash-free sessions at or above 99.95%

SENIOR PM RULE

Never ship an experiment without its paired guardrail. If the primary conversion rate rises 5% but a guardrail trips (churn or support tickets surge), the experiment is rolled back automatically.

PART 2 · HOW PRODUCT MANAGERS TRACK METRICS DAY TO DAY

14. The AARRR framework

AARRR (Dave McClure's "pirate metrics") walks through the customer lifecycle in five stages. Each stage asks one question.

Stage	Question it asks	Sample metrics
Acquisition	Are users showing interest in our product?	App downloads, site visits, social media impressions, CAC
Activation	Are users using our product (reaching the magic moment)?	Active users, % reaching the first value event
Retention	Are users using our product repeatedly?	Daily, weekly or monthly retention; churn rate
Referral	Are users bringing in more users?	Net Promoter Score (NPS), viral coefficient, referral share
Revenue	Are users generating revenue for us?	Sales, lifetime value (LTV)

What each stage teaches

- **Acquisition** happens when a customer is acquired through ads, SEO, content, sales, affiliates, community or referrals. Customers move through the AIDA funnel (awareness, interest, desire, action). High acquisition with poor retention is a leaky bucket. Growth must be steady because some users always leave.
- **Activation** happens when a customer has used the product and started deriving value. Products deliberately push users to the magic moment. Twitter suggested popular accounts after users followed 30 people. Dropbox encouraged users to upload a file during sign-up. If acquisition is broad and untargeted, most users never activate.
- **Retention** requires the customer to repeat the key value-driving action over time. Habit-forming products retain best. Users who churned can often be reactivated.
- **Referral** happens when a customer recommends the product. Organic referral chains keep acquisition costs low. Products create viral loops through referral discounts, contact-list invites and shareable content.
- **Revenue** depends on the right business model. Lifetime value must be much greater than customer acquisition cost (LTV ≫ CAC) for the business to be sustainable.

AARRR applied: Cred (credit card bill payments)

Stage	Metric
Acquisition	# of app installs
Activation	# of users who pay their first credit card bill, or set up their cards on Cred
Retention	# of users who pay n consecutive credit card bills
Referral	# of users who share rewards on social media, or sign up with a referral code or link
Revenue	# of users who buy products or services using Cred rewards

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15. Retention and product-market fit

Long-term retention is one of the best signs of product-market fit (PMF). Retention curves make it visible.

Two products are shown on the same axes over 20 time units. **Product A** drops sharply at first, then flattens into a stable tail. It has product-market fit. **Product B** keeps falling in a straight line and never flattens. It does not have

product-market fit, however large its early numbers look.

Reading the shape of a retention curve

Curve shape	What it means
Almost zero drop-off (95 of 100 users still there in month 12)	A strong habit or core utility. The retention curve looks like a flat horizontal line.
Loses 80% in the first 2 months, then the remaining 20% stay for years	A steep cliff followed by a long, stable tail. Common in products with a clear core use case.
Flat for a few months, then drops, then picks up and stays flat	Resurrection: users return later, often after a new feature or re-engagement campaign.
Constant downward diagonal (50%, 25%, 10%, then zero)	The product never flattens. No product-market fit yet.

PRACTICAL TEST

Plot retention by weekly cohort. If the curve flattens at a healthy level, you have a base to grow from. If it does not flatten, fix retention before you spend on acquisition, because the leaking bucket will waste the spend.

16. Growth loops beyond the funnel

AARRR is a linear funnel. Linear input gives linear output, and growth stops when spend stops. A compounding growth loop feeds itself: active users trigger the acquisition and activation of the next cohort.

Loop	How it works	Examples	Health metric
Viral and collaboration	A user creates an artifact, shares it, and invitees experience the product's value and sign up for their own workspace	Figma, Slack, Miro, Notion	Viral coefficient K greater than 1.0; loop cycle time
Content and SEO (user-generated)	Active users post public questions or answers; search engines index the pages; new searchers land and sign up	Stack Overflow, Quora, Pinterest, Glassdoor	Crawl-to-index ratio; UGC per active user
Paid reinvestment	Capital acquires a user; the user converts; gross margin from lifetime value funds the next acquisition	Duolingo, Uber, Wealthfront	LTV:CAC above 3.0; CAC payback under 12 months

PART 2 · HOW PRODUCT MANAGERS TRACK METRICS DAY TO DAY

17. HEART framework

HEART (from Google) is a user-experience framework. It organises metrics around five questions. Use it when AARRR is too focused on the funnel and you need to measure the quality of the experience.

Dimension	Question	Cred example
Happiness	Are users happy with the product?	App store rating
Engagement	Are users using the product deeply?	# of users who paid at least one bill and used at least one reward
Adoption	Are users starting to use the product?	# of users who pay their first credit card bill

Retention	Are users coming back repeatedly?	# of users who pay 3 consecutive credit card bills
Task success	Are users achieving their goals?	# of users who avoid late payments on bills

HEART VS AARRR

AARRR tells you where users are in their lifecycle. HEART tells you whether the experience is good. Many teams use both: AARRR to track the funnel, HEART to judge quality per feature.

18. OKRs and the metric cascade

OBJECTIVES

Visionary goals or north stars that set a clear direction. They are significant, concrete, action-oriented and inspirational. Example (Zoom): provide video experience excellence that makes every meeting feel like in-person.

KEY RESULTS

Time-bound, measurable outcomes that track progress towards the objective. They are specific, aggressive yet realistic, measurable and verifiable. Example (Zoom): reduce meeting disconnections by 30% for all calls in FY26.

How OKRs plan the work

An objective sits at the team level. Key results (KR1, KR2, KR3) measure it. Initiatives deliver the key results. Each level has a scope: the objective is set by the higher team and the key results belong to the team.

Spotify example: from vision to sprint

Vision: unlock the potential of human creativity by giving artists the chance to live off their art and fans the chance to enjoy it.

Objective	Key result (team)	Initiative
Support 10,000 new artists launching their first album in FY26	Decrease average time to launch an album by 30% (creator tools team)	Launch an AI-based creator tools suite
Generate $5M in artist revenue in FY26	Diversify up to 10% of artist revenue (new initiatives team)	Launch virtual concerts
Reach 1B monthly active users in FY26	Get 100M new users (growth team)	Launch v2 of Spotify Wrapped
Reach 10B hours of monthly streaming in FY26	Increase average session length by 20% (recommendations team)	Improve the recommendation engine using a better understanding of song vibes

The execution bridge runs from a long-term North Star (total monthly listening hours), through input drivers (discovery breadth, personalisation depth, social-sharing velocity), to team-level OKRs (for example, the search team's goal of an immediate, skip-free magic track, with a key result to lift day-7 skip-free playback from 68% to 82%), and finally to bi-weekly sprints and experiments.

PART 2 · HOW PRODUCT MANAGERS TRACK METRICS DAY TO DAY

19. Analytics tools hierarchy

Think of analytics as a pyramid. Each layer depends on the one below it.

Layer (bottom to top)	What happens there
1. Data collection	Events and properties are captured reliably (basic requirement)
2. Data warehousing and pipelining	Data is stored and moved into a usable shape (basic requirement)

3. Data analysis	Questions are answered with segments, funnels and cohorts. **This is where most teams get stuck.** A product analytics tool helps most here.
4. Insight generation	Patterns become explanations and recommendations
5. Decision automation	Decisions run themselves, for example automatic rollbacks. Few teams do this.

The class recommended a Mixpanel tutorial for the analysis layer, and an external overview of the product analytics market.

20. Metrics cheat sheet

Business metrics

- **Revenue:** ARPU, lifetime value (LTV), annual recurring revenue (ARR), gross merchandise value (GMV), average order value (AOV)
- **Costs:** customer acquisition cost (CAC), cost of goods sold (COGS), marketing costs, fixed and variable costs
- **Profit:** gross margin, net margin, unit contribution

Product metrics

- **Acquisition:** app installs, downloads, sign-ups, unique visitors, bounce rate
- **Activation:** MAU / WAU / DAU, session count, conversion rate
- **Engagement:** engaged users (MEU / WEU / DEU), session length, feature usage
- **Retention:** week-over-week (WoW), month-over-month (MoM), week-4 retention, churn rate
- **Satisfaction:** CSAT, NPS, app store rating, support tickets, bugs

Feature metrics

- Adoption, discovery (first-time adopters), repeat adopters
- Usage funnel, usage heatmap, task completion
- **Efficacy:** task success rate, task failure rate
- **Efficiency:** time to complete, steps to complete, cognitive load

Engineering metrics

- **Latency:** page load time (PLT), time to interactive (TTI), visually complete time (VCT), cold and warm start time, end-to-end (E2E) action latency, perceived latency
- **Reliability:** E2E action reliability, API reliability, mean time to respond (MTTR), mean time between failures (MTBF), downtime, API throttling rate

PART 2 · HOW PRODUCT MANAGERS TRACK METRICS DAY TO DAY

21. Interview answer template

When an interviewer asks "How would you measure success for X?", answer in this order. It takes about two minutes and shows structured thinking.

1. **Users and stakeholders.** Name each user type and its JTBD.
2. **North Star.** One metric that captures value delivered to the main user.
3. **Ladder.** Two or three inputs (leading, controllable) and the outputs they drive (lagging, business).
4. **Business metric.** How the product supports revenue or the business model.
5. **Guardrails.** One counter-metric per primary metric, for example retention, uninstalls, latency or trust.
6. **Diagnostics.** How you would find the cause if the North Star moved (funnel steps, cohorts, segments).

Common mistakes to avoid

- Choosing a vanity metric (total sign-ups, total downloads) that does not reflect value.
- Reporting a metric without a time grain or cohort.
- Optimising one metric without a guardrail, which invites Goodhart's Law.
- Measuring outputs only. Leading inputs are what the team can act on.

- Using averages to judge retention. Always look at cohorts.
- Declaring product-market fit from acquisition numbers. Check whether the retention curve flattens.

MODULE SUMMARY

Define success for each stakeholder (JTBD). Pick a North Star. Build a ladder of leading inputs and lagging outputs. Add guardrails to every primary metric. Track the lifecycle with AARRR and quality with HEART. Check retention curves for product-market fit. Connect the metrics to OKRs so the team can act on them.

Source material: Week 6 Analytics and Metrics class (Product Management Launchpad). Full class slides are available as a separate PDF. The quick-notes PDF lists the key pointers for revision.